Quality & Production Management Dashboard — Dairy Packaging Line

Tool: Microsoft Power BI | Domain: Food Tech / QC / Production Analytics

Project Objective

Design a management dashboard to jointly monitor product quality (QC) and production line performance in a dairy plant, focused on industry-standard KPIs (OEE, First Pass Yield, Cpk, downtime Pareto) so management can assess line status and make decisions at a glance.

Data Source

The dataset used is modeled on the structure, value ranges, and labeling logic of three public, redistributable Kaggle datasets:

- Milk Quality Prediction — lab test parameters for milk (pH, temperature, fat, turbidity, colour, taste, odor)
- Production Quality (Sensor Dataset) — temperature and humidity sensors on a production line
- Manufacturing Downtime / Efficiency Dataset — downtime events and their causes

Since no dairy plant publishes its internal data publicly, the data for this project was synthetically generated within realistic industry ranges so it stays valid for analytical practice; the full data-generation logic is documented in the attached Excel file.

Key Performance Indicators (KPIs)

KPI	Formula	Purpose
OEE	Availability × Performance × Quality	Overall equipment effectiveness of the line
First Pass Yield	Grade=High batches ÷ total batches	Share of product made right the first time
Rejection Rate	Grade=Low batches ÷ total batches	Quality rejection rate
Out-of-Spec Rate	Batches outside allowed range ÷ total	Monitoring quality parameter deviation
Downtime Pareto	Share and cumulative % of each downtime cause	Prioritizing corrective actions

Key Findings

- The night shift shows a lower First Pass Yield and higher rejection rate than the morning and afternoon shifts; the main driver is elevated raw-milk storage temperature.
- “Mechanical failure” (٢٧٪) and “CIP / cleaning” (٢٤٪) together account for more than half of total line downtime.
- Availability has remained stable and above ٩٠٪ throughout the review period.

Recommended Actions

- Implement a preventive maintenance (PM) program on the equipment with the highest mechanical failure rate.
- Review the CIP cleaning schedule to reduce downtime without compromising hygiene standards.
- Tighten cold-chain temperature control on the night shift to bring quality in line with other shifts.

Tools & Method

Power BI Desktop for data modeling (Star Schema), DAX calculations, and visual design; Microsoft Excel for raw data preparation and documentation.

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